General Biology 1

First Semester

Supplementary Materials

Module 6

Cell Division

(Eukaryotic vs Prokaryotic)

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What I Know

Directions: Read each situation carefully and choose the best answer.

1. Which statement best describes cell division in prokaryotic cells?

a. It involves mitosis with spindle fiber formation
b. It occurs through binary fission without a nucleus
c. It includes multiple phases such as prophase and metaphase
d. It requires a nuclear membrane to break down

2. What is the key difference between binary fission and mitosis?

a. Binary fission produces genetically different cells
b. Mitosis occurs only in bacteria
c. Binary fission is simpler and does not involve spindle fibers
d. Mitosis does not involve DNA replication

3. During eukaryotic cell division, which structure ensures proper separation of chromosomes?

a. Ribosomes
b. Spindle fibers
c. Cell membrane
d. Cytoplasm

4. A student observes a cell dividing without visible stages like metaphase or anaphase. What type of cell is most likely being observed?

a. Animal cell
b. Plant cell
c. Eukaryotic cell
d. Prokaryotic cell

5. Which of the following is common to both prokaryotic and eukaryotic cell division?

a. Formation of spindle fibers
b. Presence of a nucleus
c. DNA replication before division
d. Multiple phases of mitosis

What I Need to Know

This supplementary module is designed to help you understand the difference between the **cell division of eukaryotic and prokaryotic cell**. In this lesson, you will explore how the key feature and process occurring in each stage of prokaryotic and eukaryotic cell division.

Lesson 6 Cell Division (Eukaryotic vs Prokaryotic)

Cell division is the process by which cells reproduce to ensure growth, repair, and continuity of life. In prokaryotic cells, such as bacteria, division occurs through a simple process called binary fission, while in eukaryotic cells it undergoes a more complex process involving mitosis and cytokinesis. both processes result in genetically identical cells, eukaryotic cell division is more regulated and occurs in distinct stages, reflecting the greater complexity of eukaryotic cells compared to prokaryotes.

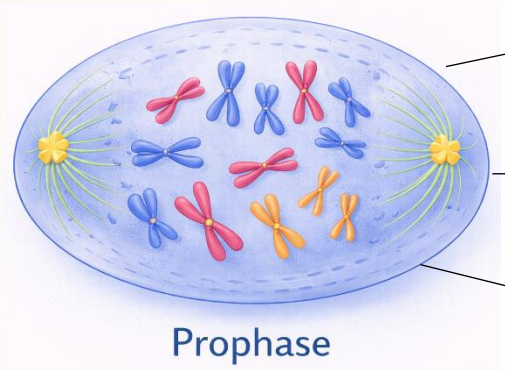
After going through this supplementary module, you are expected to:

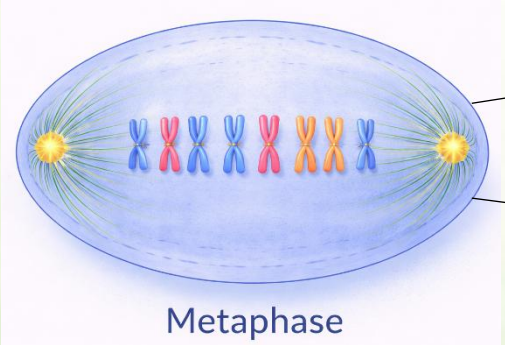
1. identify the stages of cell division in prokaryotic cells (binary fission) and eukaryotic cells (mitosis and cytokinesis);
2. describe the key features and processes occurring in each stage of prokaryotic and eukaryotic cell division; and
3. compare and differentiate prokaryotic and eukaryotic cell division in terms of stages, structures involved, and overall process.

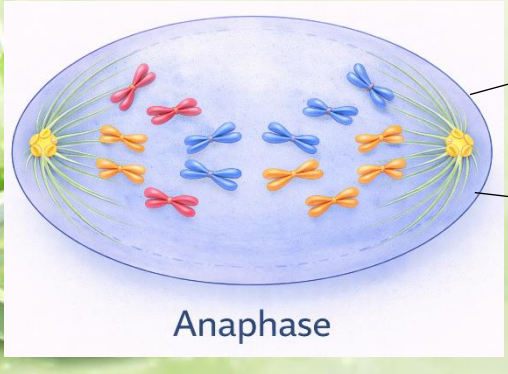
What's In

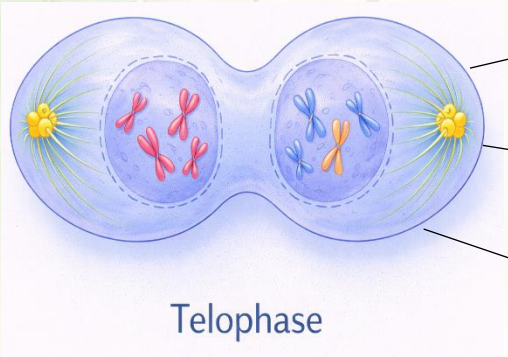
Activity 1: "Divide and Conquer: The Stages of Mitosis"

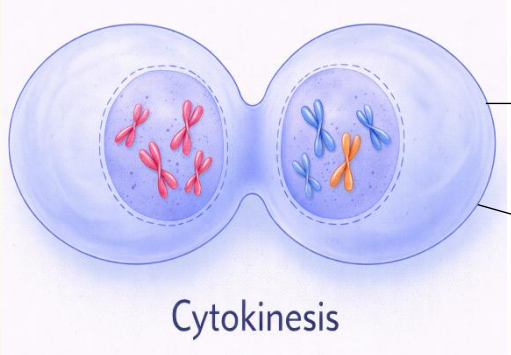
Directions: Identify in which phase of mitosis the series of event happened.





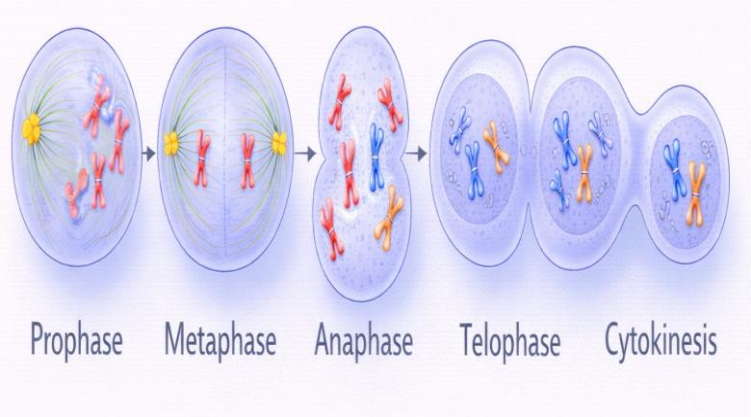






EVENTS IN MITOSIS

Nuclear membrane begins to break down	Cell membrane pinches (animal cells) or cell plate forms (plant cells)
Sister chromatids separate at the centromere	Spindle fibers start to form
Chromosomes begin to decondense	Chromosomes align at the center of the cell
Cytoplasm divides into two daughter cells	Nuclear membranes reform around each set of chromosomes
Chromosomes condense and become visible	Spindle fibers attach to the centromeres of chromosomes
Chromatids are pulled toward opposite poles	Chromosomes reach opposite poles



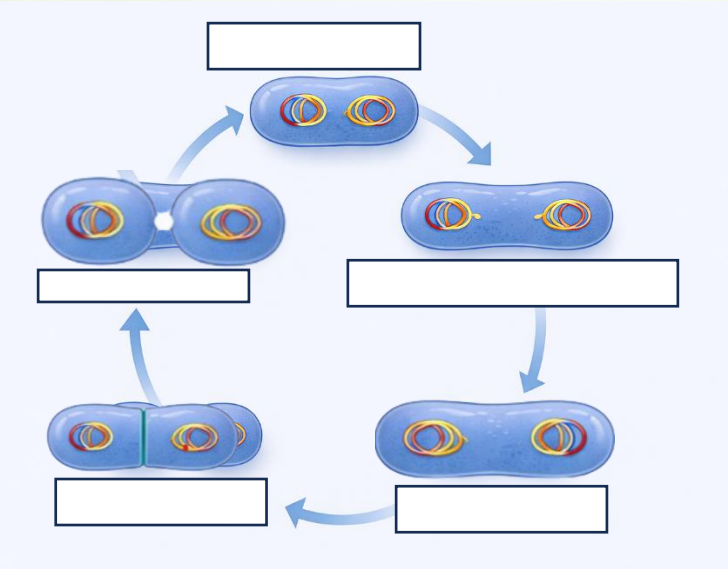
Mitosis is the process by which a eukaryotic cell divides its nucleus to produce two identical sets of chromosomes. It ensures that each daughter cell receives the same genetic information. Mitosis occurs in a series of organized phases: **prophase**, **metaphase**, **anaphase**, **telophase**, followed by **cytokinesis.**

*(To know more about these phases of mitosis, refer to the information in **Page 40** of this supplementary module.)*

What's New

Activity 2: "Split to Survive: The Binary Fission Cycle"

Directions: Complete the stages of cell division in prokaryotic cell. Cut and paste it correctly.



DNA moves to opposite ends

Cell splits into two

DNA replication

Cell elongation

Septum formation

Binary fission is the process by which **prokaryotic cells** (such as bacteria) reproduce. It is a simple, rapid form of cell division that allows one cell to divide into two genetically identical daughter cells. Unlike

eukaryotic cell division, binary fission does not involve a nucleus or complex stages like mitosis. Instead, it follows a direct and efficient sequence of events:

The process begins with **DNA replication**, where the single circular chromosome of the prokaryotic cell is copied. This results in two identical DNA molecules, ensuring that each future daughter cell will receive the same genetic information.

Next is **cell elongation**, during which the cell increases in size. As the cell grows longer, the two copies of DNA begin to move apart within the cytoplasm. This prepares the cell for proper separation of genetic material.

In the third stage, **DNA moves to opposite ends of the cell**. Each copy of the chromosome attaches to different parts of the cell membrane and is actively separated as the cell elongates. This ensures that both sides of the cell contain one complete set of DNA.

Following this is **septum formation**, where the cell membrane begins to pinch inward at the center. A structure called the septum forms, gradually dividing the cell into two compartments. This step marks the beginning of physical separation.

Finally, the cell undergoes **cell division**, where the septum fully forms and the cell splits into two identical daughter cells. Each new cell contains one copy of the original DNA and can continue the cycle.

Overall, binary fission is an efficient and continuous cycle that enables prokaryotic organisms to reproduce quickly, often leading to rapid population growth under favorable conditions.

What's More

Activity 3: "One Cell, Two Systems: Comparing Life's Division Strategies"

Directions: Using what you have learned in the previous activity, complete the comparison table about the difference between cell division of Prokaryotic and Eukaryotic Cell.

Feature	Prokaryotic Cell Division	Eukaryotic Cell Division
Type of Process		
Stages		
Structures Involved		
Complexity		

Binary fission

Mitosis

DNA replication → Cell elongation → DNA moves to opposite ends → Septum formation → Cell splits into two

Prophase → Metaphase → Anaphase → Telophase → Cytokinesis

Nucleus, chromosomes, spindle fibers, centrosomes, cell membrane

Circular DNA, cell membrane, cytoplasm

Simple and rapid process

More complex and highly regulated process

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A scientist observes a cell dividing rapidly without visible stages such as metaphase or anaphase. Which conclusion is most accurate?

   a. The cell is undergoing mitosis in a eukaryote
   b. The cell is undergoing binary fission in a prokaryote
   c. The cell skipped the G_2 phase
   d. The cell is undergoing meiosis

2. Why do eukaryotic cells require spindle fibers during cell division, while prokaryotic cells do not?

   a. Prokaryotic cells have multiple chromosomes
   b. Eukaryotic cells lack DNA
   c. Eukaryotic cells need to organize and separate chromosomes within a nucleus
   d. Prokaryotic cells divide more slowly

3. A mutation prevents the formation of spindle fibers in a cell. Which type of cell division would be directly affected?
   a. Binary fission in prokaryotes
   b. Mitosis in eukaryotes
   c. DNA replication in both cell types
   d. Cytokinesis in prokaryotes
4. Which scenario best explains why prokaryotic cells can reproduce faster than eukaryotic cells?
   a. Prokaryotic cells have more complex structures
   b. Prokaryotic cells undergo multiple phases of mitosis
   c. Prokaryotic cells use a simpler division process without nucleus or spindle fibers
   d. Eukaryotic cells do not replicate DNA
5. A student claims that prokaryotic and eukaryotic cell division are identical because both produce two cells. Which statement best evaluates this claim?
   a. Correct, because both processes use spindle fibers
   b. Incorrect, because only eukaryotes replicate DNA
   c. Incorrect, because the processes differ in stages, structures, and complexity
   d. Correct, because both processes involve mitosis